

CPCS-241 Midterm Exam 1st term 2022-2023

PART 1: Multiple Choice Questions

Q1: Program-data-independence in database management systems is achieved through_____ .

- a) the three-schema architecture
- b) The internal DB schema
- c) The external DB schema
- d) The DBMS catalog

Q2: The self-describing nature of the database systems is achieved through_____ .

- a) The three-schema architecture
- b) The internal DB schema
- c) The external DB schema
- d) The DBMS catalog

Q3: The _____ describes how data is stored as files in the computer by representing info such as record formats, record orderings, and access paths (indexes on attributes).

- a) Conceptual (semantic) data model
- b) Physical (internal) data model
- c) Implementation (representational) data model
- d) External(view) data model

Q4: Which one of the following is NOT a typical functionality of the DBMS?

- a) Defining the database in terms of its data types, structures, and constraints
- b) Manipulating the database through insertions, deletions, and updates
- c) Developing applications in a high-level language
- d) Loading the initial database contents on secondary storage

Q5: In an ER-diagram, if each project has a unique name, then the name attribute is modeled to as a _____ .

- a) Primary key attribute
- b) An atomic attribute
- c) Key attribute
- d) Single valued attribute

Q6: The database state which satisfies the constraints, and the specified schema structure is called:

- a) Intention state
- b) Extension state
- c) Valid state
- d) Initial state

Q7: One of the advantages of using the database approach is the sharing of data and multi-user transaction processing.

- a) True
- b) False

Q8: System protection against software and hardware crashes is the responsibility of the DBMS.

- a) True
- b) False

Q9: Physical data independence means the capacity to change the conceptual schema without having to change the external schemas and their associated application programs.

- a) True
- b) False

Q10: Inserting a new record into a DB table could violate the domain constraint.

- a) True
- b) False

Q11: Entity integrity constraint involves two relations and is used to specify a relationship among tuples in two relations.

- a) True
- b) False

Q12: Answer the following questions:

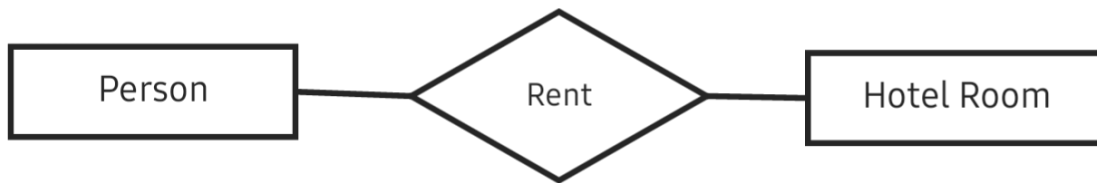
Q12.a: In the senior projects database, consider the following business rules:

- A student may be the leader of only one group, and each group should be led by only
- Not all students lead groups, but all groups must be led by a student.

What type of relationship is this?

- a) One-to-one with full participation of group entity
- b) One-to-one with full participation of student entity
- c) Many-to-one with full participation of group entity
- d) di Many-to-many with full participation of student entity

Q12.b: Consider the following ER diagram for a hotel:



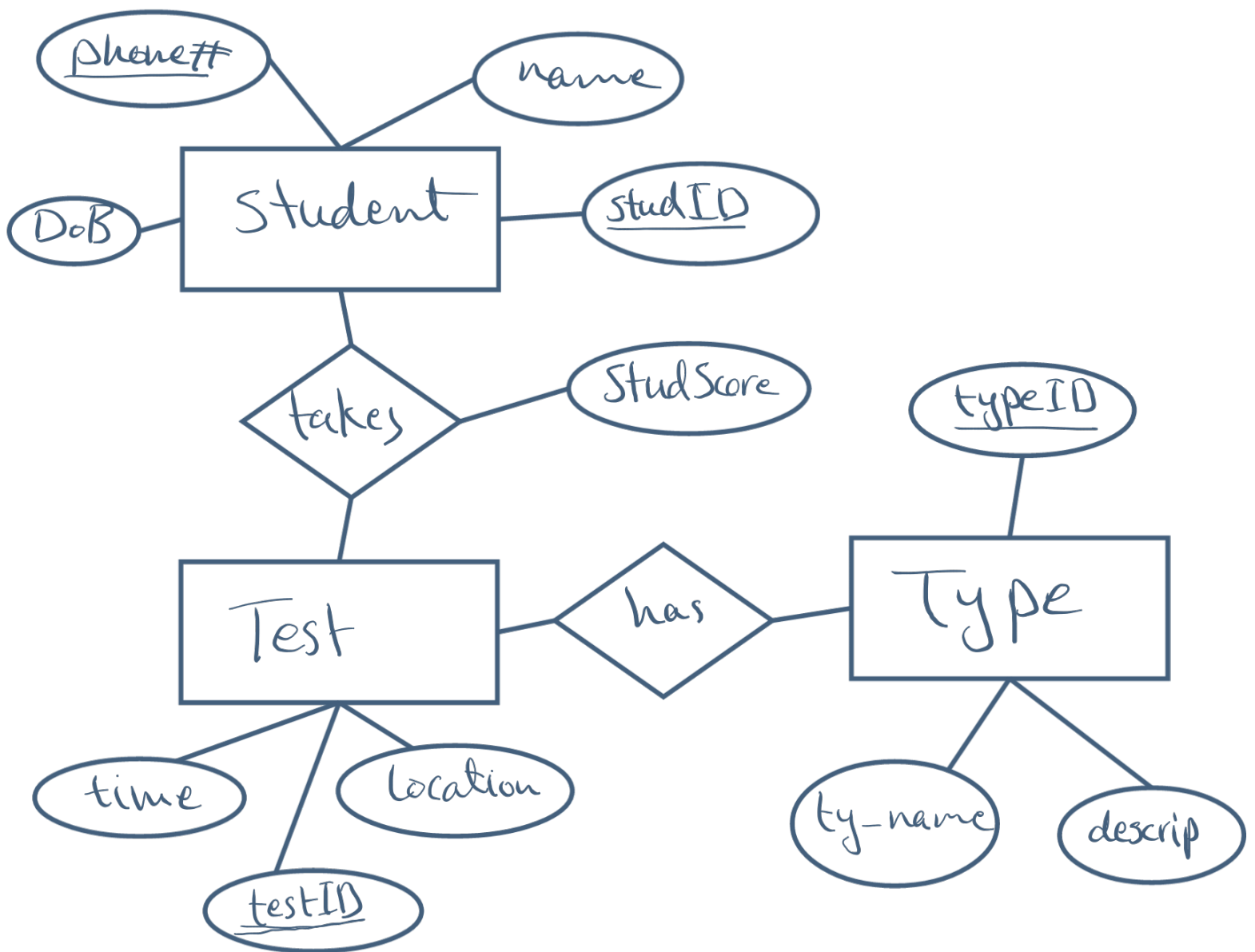
Suppose we want to keep track of the rent price each time a person rents a room in a hotel, then this information should be an attribute of _____ .

- a) Person
- b) Hotel room
- c) Rent

Q13: Draw the ER Diagram for the Standardized Exams database, which keeps track of scored standardized tests (e.g., TOEFL, IELTS, SAT, etc) that students take. Consider the following description for drawing the ER diagram:

- Each student has a student ID, Name, date of birth, and phone number.
- Each test has a test ID, type, time, and location.
- A test type consists of type ID, type name, and description of the purpose of the test.
- A student can take more than one test, and each time the student takes the test, his/her score is recorded in the Database
- Multiple students can register and take a test.

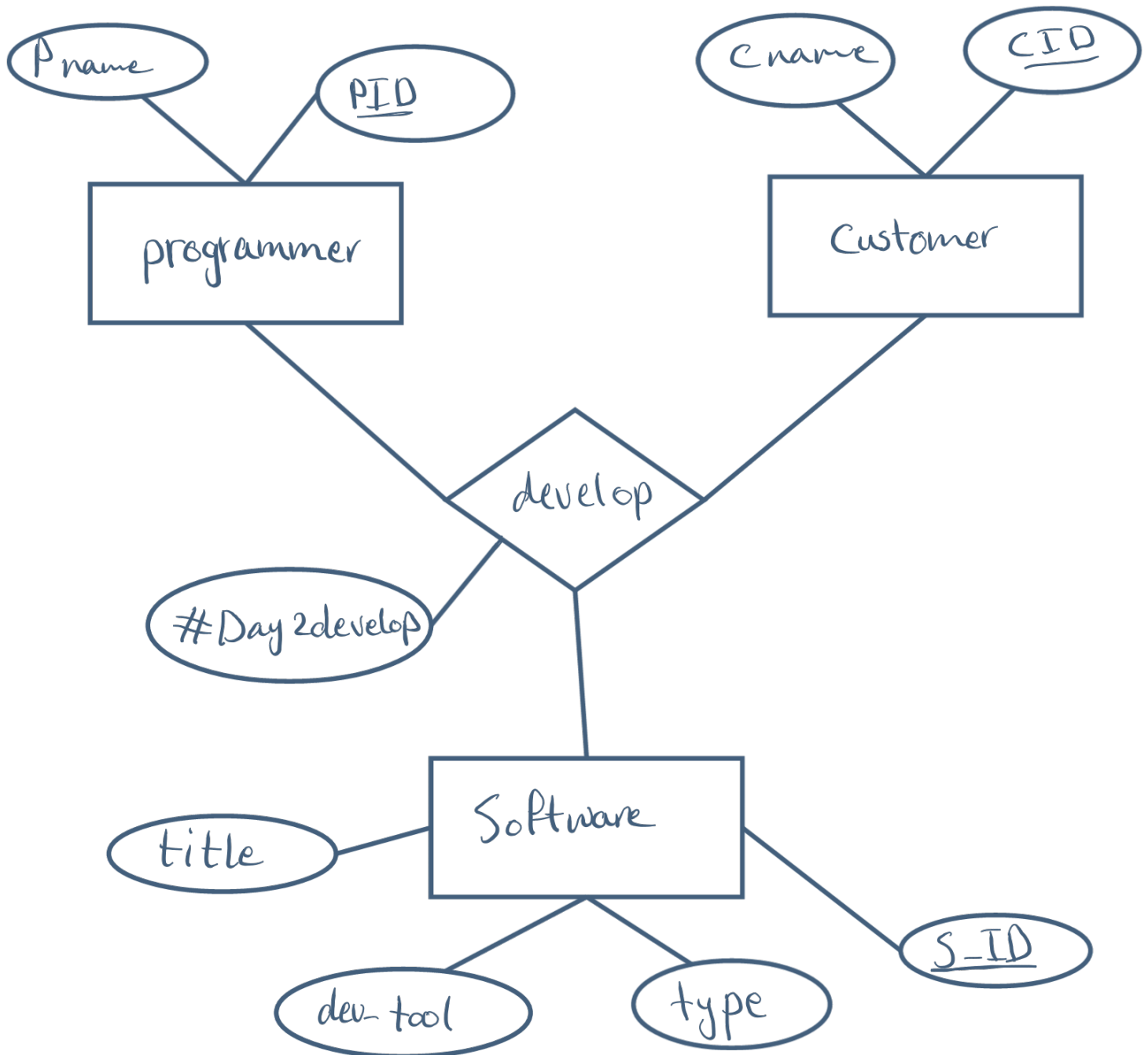
Show entities, attributes, relationships, keys, and cardinality ratios on your diagram



Q14: Draw the ER diagram for the following description of a database for a Software Development Company, which keeps track of the programmers, customers, and software.

- A customer has an ID, and name.
- A programmer has an ID, and name.
- A software has an ID, title, type (e.g. application or website), and the development tool used to develop the software.
- A programmer develops a particular software for a particular customer, and the number days needed for development is stored in the database.

Show entities, attributes, relationships, keys on your diagram

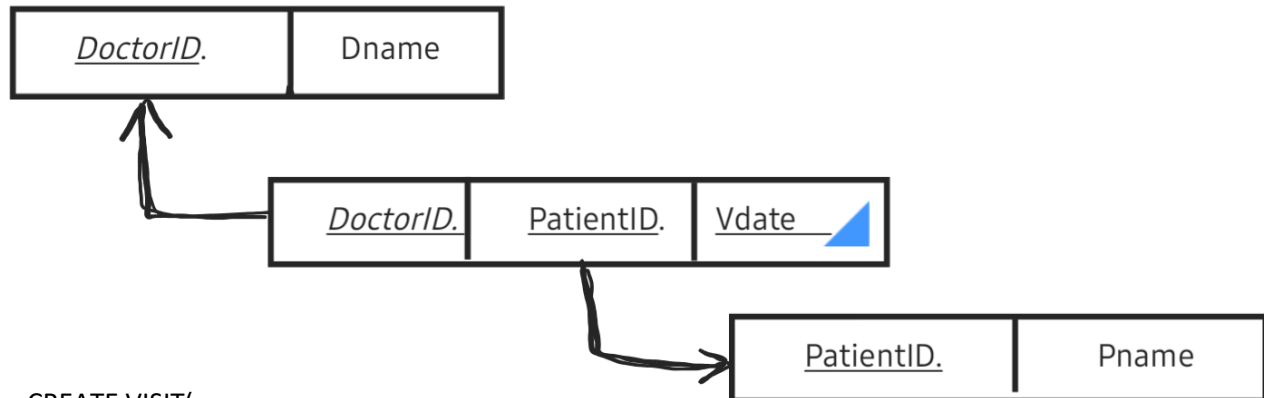


Q15: Given the following Clinic DB and the SQL script for VISIT table, answer questions Q15.a and Q15.b.

DoctorID	Dname
D001	Lara
D002	Mohamed
D003	Nour
D004	Yaser

PaitientID	Pname
P001	Dana
P002	Ahmed
P003	Ali
P004	Fay

DoctorID	PaitientID	Vdate
D001	P003	12-1-2021
D002	P003	
D003	P004	
D004	P004	



```

CREATE VISIT(
FOREIGN KEY (DoctorID) REFERENCES DOCTOR (DoctorID);
FOREIGN KEY (PatientID) REFERENCES PATIENT (PatientID) ON UPDATE CASCADE; )
  
```

Q15.a: What would happen when this operation is executed?

In VISIT table, update the PatientID to 'P004' where doctorID ='D001' and vdate =12-1-2021

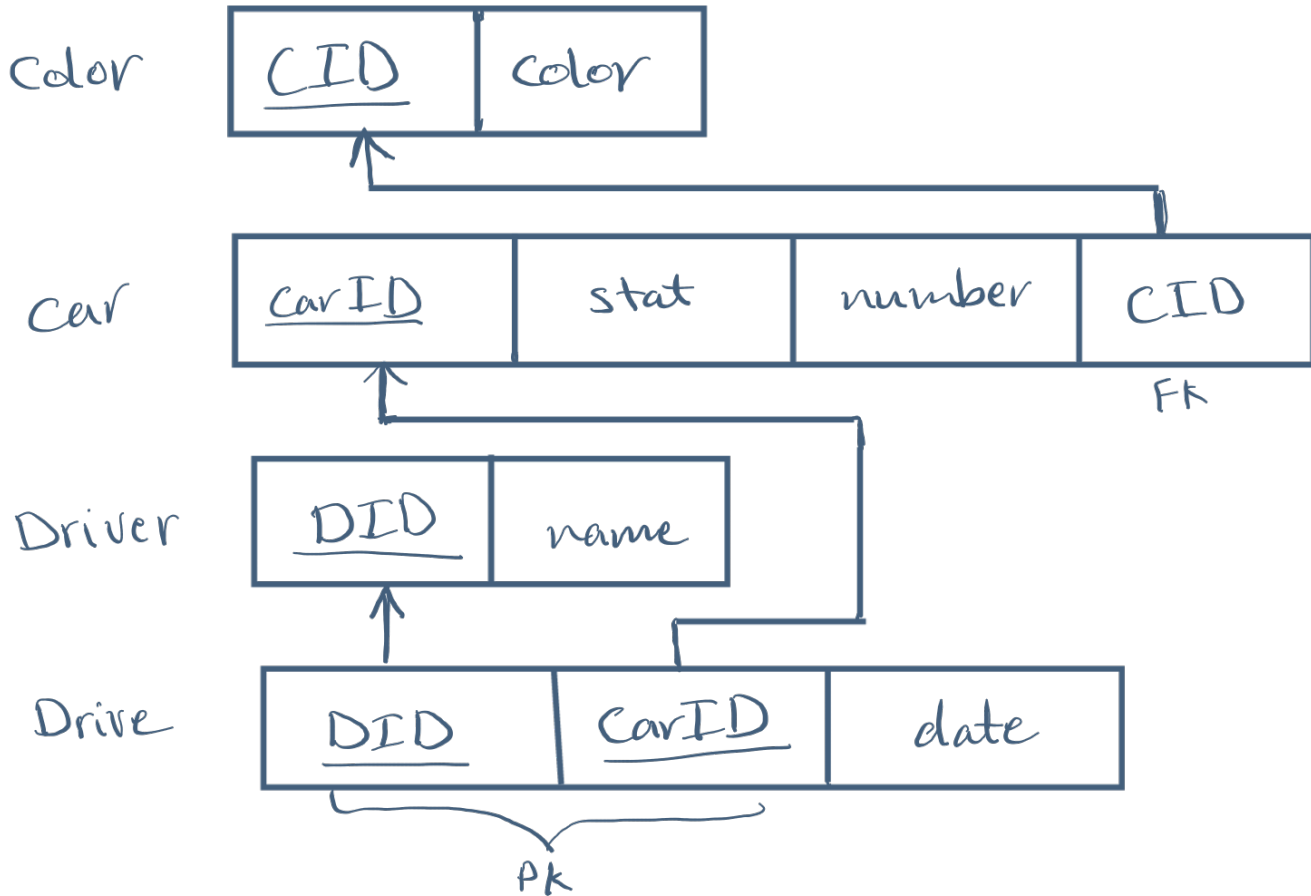
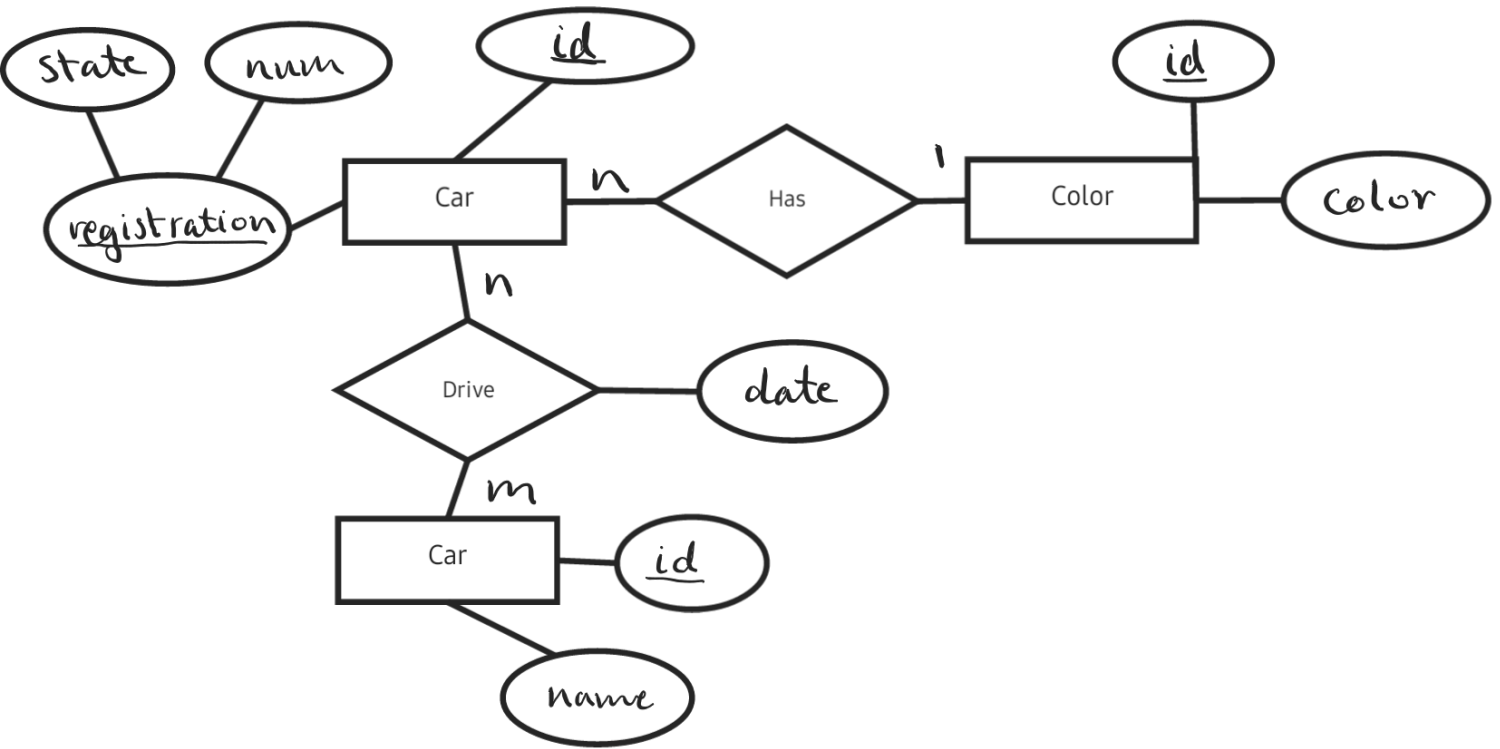
- a) The operation will be successfully executed, and the VISIT table will be updated.
- b) The operation will be rejected because it violates the key constraints
- c) The operation will be rejected because it violates the entity integrity constraint.
- d) The operation will be rejected because it violates the referential integrity constraints.

Q15.b What would happen when this operation is executed?

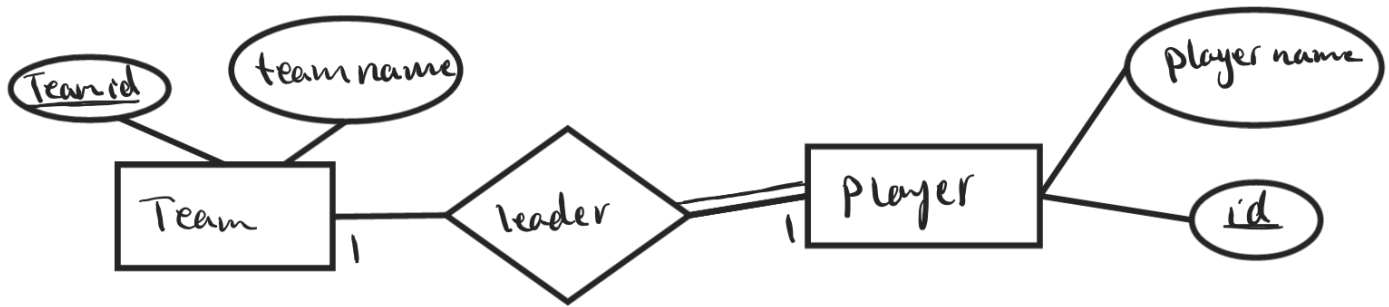
Delete from DOCTOR the tuple with DoctorID='D001'

- a) The operation will be successfully executed, and the DOCTOR table will be changed.
- b) The operation will be rejected because it violates the key constraints.
- c) The operation will be rejected because it violates the entity integrity constraint.
- d) The operation will be rejected because it violates the referential integrity constraints.

Q16: Map the ER to schema diagram

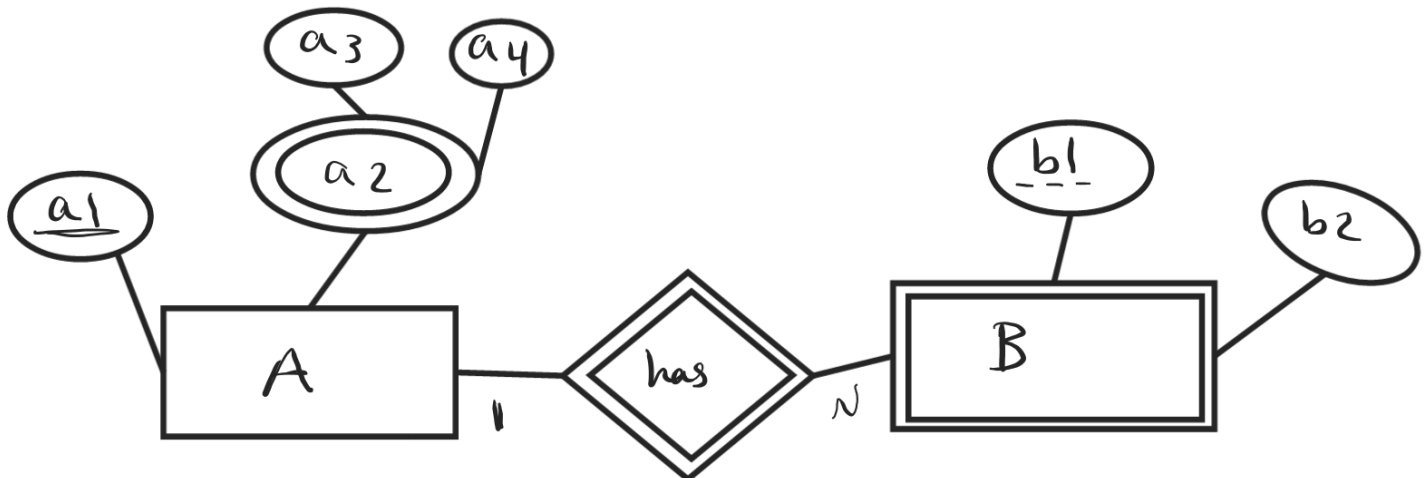


Q17: Chose the correct mapping for the following ER diagram:



- a) Foreign key approach, add the primary key of PLAYER as a foreign key in TEAM.
- b) Foreign key approach, add the primary key of TEAM as a foreign key in PLAYER
- c) Merge the TEAM and PLAYER in one relation TEAM PLAYER

Q18 : Consider the following ER diagram.



Which one of the following schema diagrams represents a correct mapping for the ER diagram?

